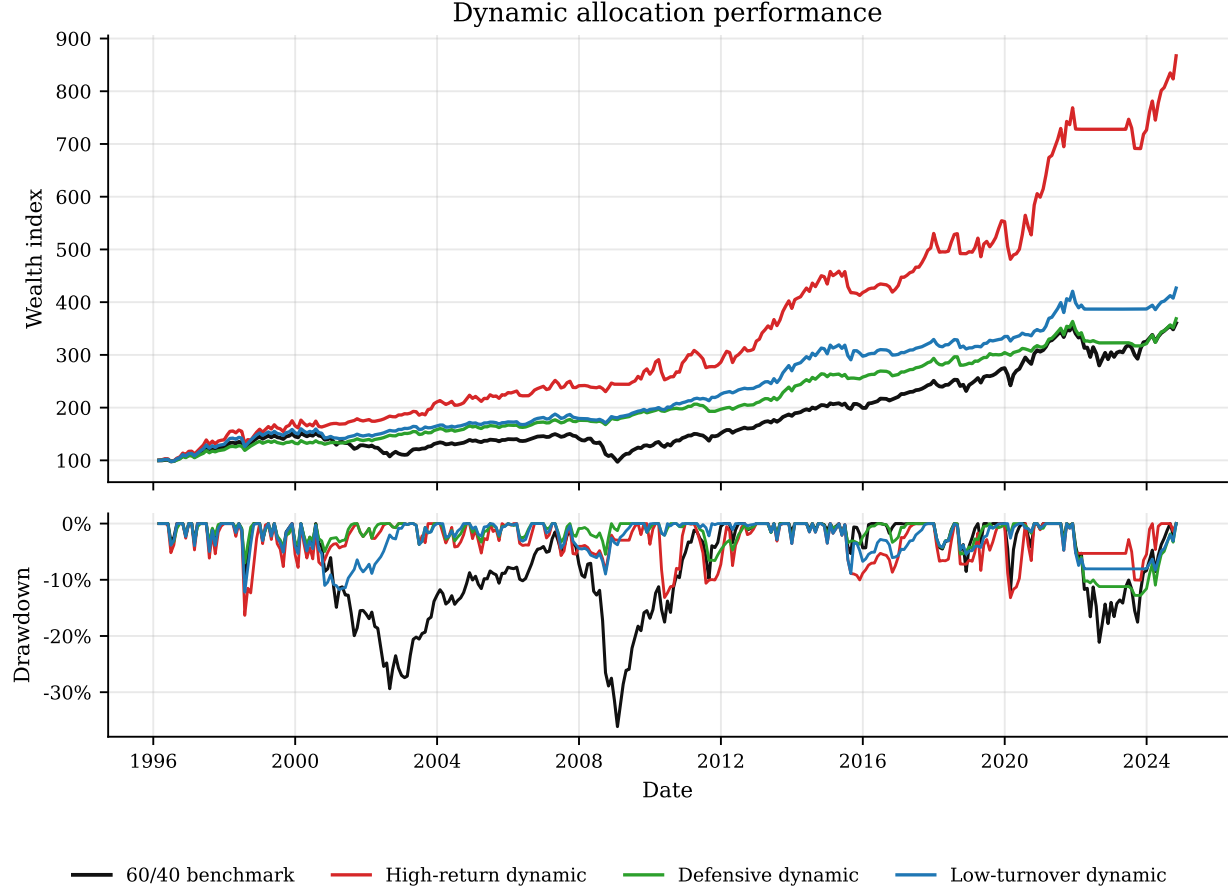


Table 1: Allocation performance summary

Role	Model	Predictor set	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Ann. turnover	Final wealth
60/40 benchmark	--	--	4.9%	9.6%	0.51	-36.1%	19.7%	360.37
High-return dynamic	Logistic regression	smoothed z-score predictors	8.1%	10.4%	0.77	-16.3%	182.7%	867.20
Defensive dynamic	Random forest	smoothed z-score predictors	4.7%	5.8%	0.81	-12.8%	73.9%	368.53
Low-turnover dynamic	Logistic regression	z-score + smoothed z-score predictors	5.3%	7.2%	0.74	-12.2%	57.3%	426.74

Note. Dynamic strategy returns are net of transaction costs. The 60/40 benchmark is rebalanced monthly and reported net of the same one-way transaction-cost convention. The benchmark and dynamic strategies are evaluated using the same monthly return, drawdown, turnover, and final-wealth definitions.

**Figure 1:** Dynamic allocation performance

Note. High-return dynamic is the logistic-regression strategy with smoothed z-score predictors. Defensive dynamic is the random-forest strategy with smoothed z-score predictors. Low-turnover dynamic is the logistic-regression strategy with z-score and smoothed z-score predictors.

Dynamic allocation mechanics

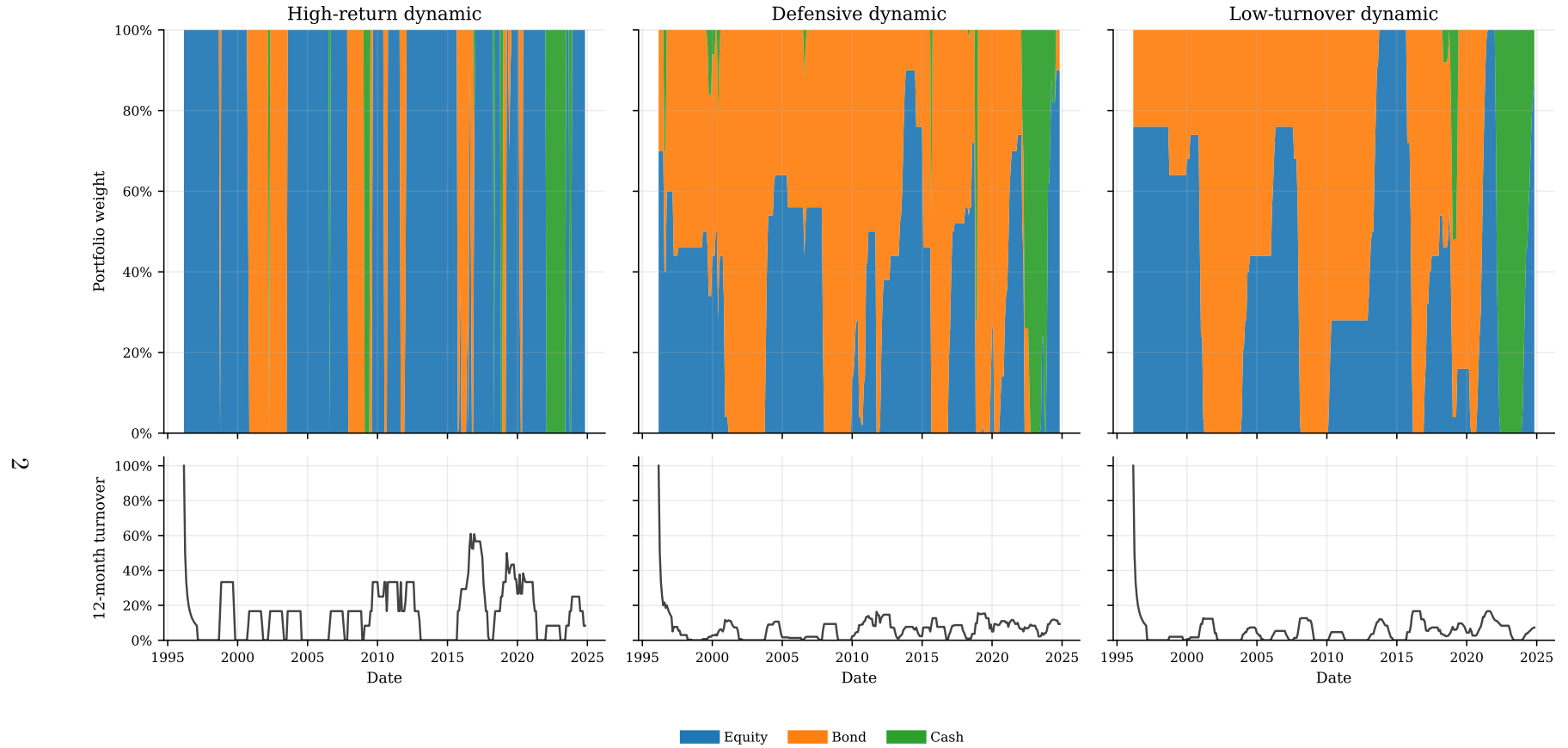


Figure 2: Dynamic allocation mechanics

Note. The columns show the high-return, defensive, and low-turnover dynamic strategies. Portfolio weights are monthly allocations to equity, bond, and cash. Turnover is the rolling 12-month average of monthly turnover.

Table 2: Selected dynamic allocation specifications

Role	Model	Predictor set	Risk aversion	Turnover penalty	Transaction cost	Smoothing	Ann. return	Sharpe	Max. drawdown	Ann. turnover	Final wealth
High-return dynamic	Logistic regression	smoothed z-score predictors	0.5	0	5 bps	0	8.1%	0.77	-16.3%	182.7%	867.20
Defensive dynamic	Random forest	smoothed z-score predictors	3	0.001	5 bps	0	4.7%	0.81	-12.8%	73.9%	368.53
Low-turnover dynamic	Logistic regression	z-score + smoothed z-score predictors	3	0.001	5 bps	3	5.3%	0.74	-12.2%	57.3%	426.74

Note. The table reports the parameter settings used for the three dynamic strategies shown in the main allocation figures.

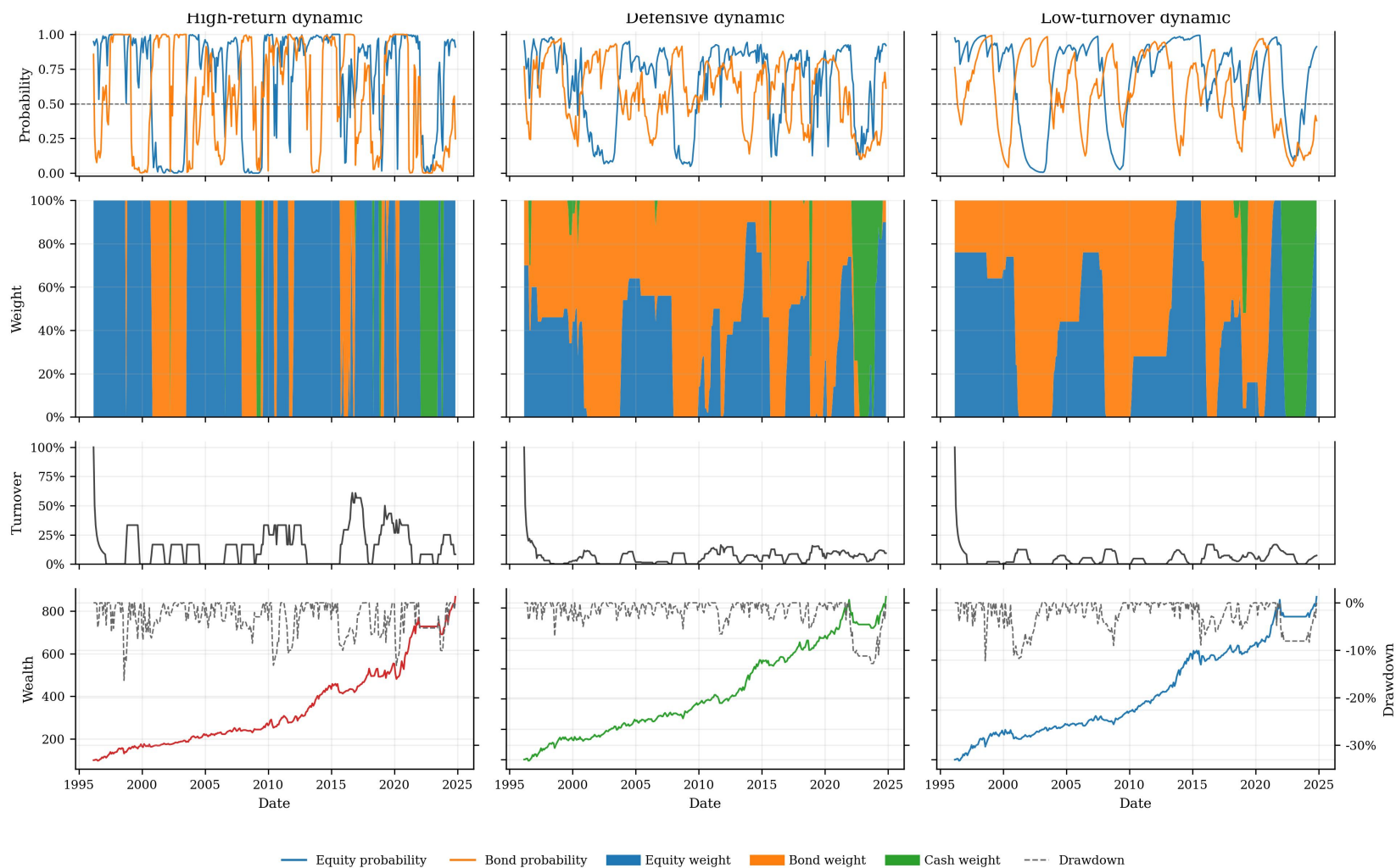


Figure 3: Allocation mechanics for selected dynamic strategies

Note. The panels report predicted favorable-regime probabilities, portfolio weights, rolling 12-month turnover, wealth, and drawdown for the three selected dynamic strategies.

Parameter sensitivity around selected strategies

The sensitivity checks vary allocation parameters around the three selected dynamic strategies. In each one-parameter sweep, the reported parameter varies and the remaining allocation parameters are fixed at the strategy-specific baseline values in Table 2. The transaction-cost and turnover-penalty table varies these two parameters jointly and keeps the remaining parameters fixed at the same baseline values.

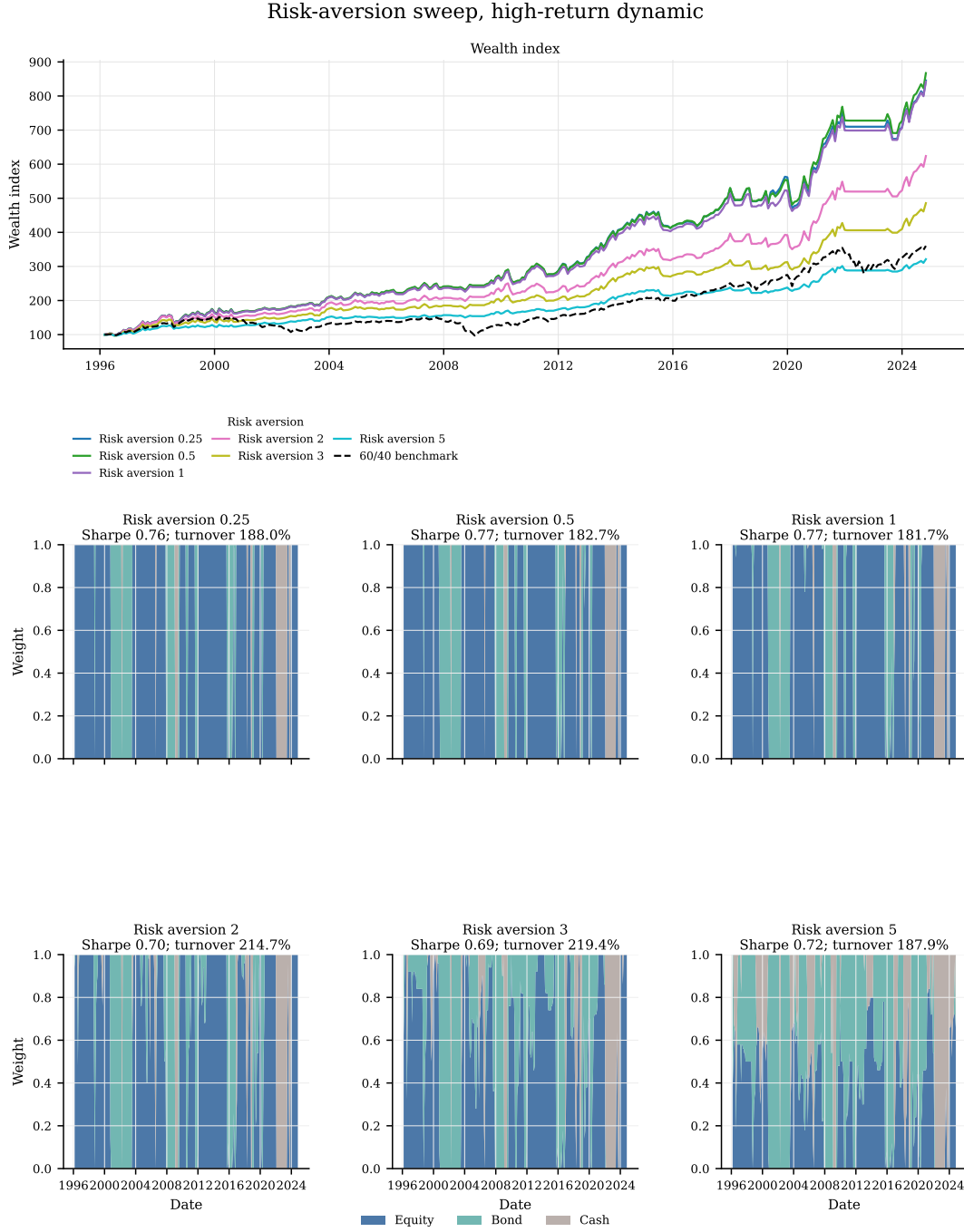


Figure 4: Risk-aversion sweep, high-return dynamic

Note. Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table 2.

Table 3: Risk-aversion sweep, high-return dynamic

Risk aversion	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0.25	8.0%	10.6%	0.76	-16.7%	-10.1%	845.74	188.0%
0.5	8.1%	10.4%	0.77	-16.3%	-9.7%	867.20	182.7%
1	8.0%	10.3%	0.77	-16.3%	-9.7%	841.89	181.7%
2	6.9%	9.9%	0.70	-16.3%	-9.7%	623.91	214.7%
3	5.9%	8.6%	0.69	-13.6%	-8.1%	485.66	219.4%
5	4.2%	5.8%	0.72	-7.7%	-5.3%	321.27	187.9%



Figure 5: Risk-aversion sweep, defensive dynamic

Note. Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table 2.

Table 4: Risk-aversion sweep, defensive dynamic

Risk aversion	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0.25	6.7%	10.6%	0.64	-19.3%	-10.2%	589.02	113.7%
0.5	6.6%	10.3%	0.64	-19.6%	-10.2%	576.23	113.9%
1	6.6%	9.9%	0.66	-17.9%	-9.7%	572.04	105.0%
2	5.7%	7.6%	0.75	-13.5%	-7.0%	473.23	94.9%
3	4.7%	5.8%	0.81	-12.8%	-5.2%	368.53	73.9%
5	3.7%	4.3%	0.84	-11.6%	-3.8%	278.55	54.8%



Figure 6: Risk-aversion sweep, low-turnover dynamic

Note. Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table 2.

Table 5: Risk-aversion sweep, low-turnover dynamic

Risk aversion	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0.25	6.5%	11.6%	0.56	-25.1%	-11.7%	536.01	71.9%
0.5	6.4%	11.3%	0.57	-23.7%	-11.0%	519.80	73.0%
1	6.6%	10.3%	0.64	-20.9%	-9.7%	563.38	68.2%
2	6.2%	9.1%	0.69	-17.7%	-9.0%	528.38	65.7%
3	5.3%	7.2%	0.74	-12.2%	-7.2%	426.74	57.3%
5	4.0%	5.1%	0.78	-8.6%	-4.7%	302.04	48.0%



Figure 7: Probability-smoothing sweep, high-return dynamic

Note. Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table 2.

Table 6: Probability-smoothing sweep, high-return dynamic

Smoothing half-life	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	8.1%	10.4%	0.77	-16.3%	-9.7%	867.20	182.7%
3	5.8%	11.1%	0.52	-23.2%	-11.5%	442.34	105.9%
6	6.4%	11.9%	0.54	-26.5%	-11.7%	514.74	59.7%

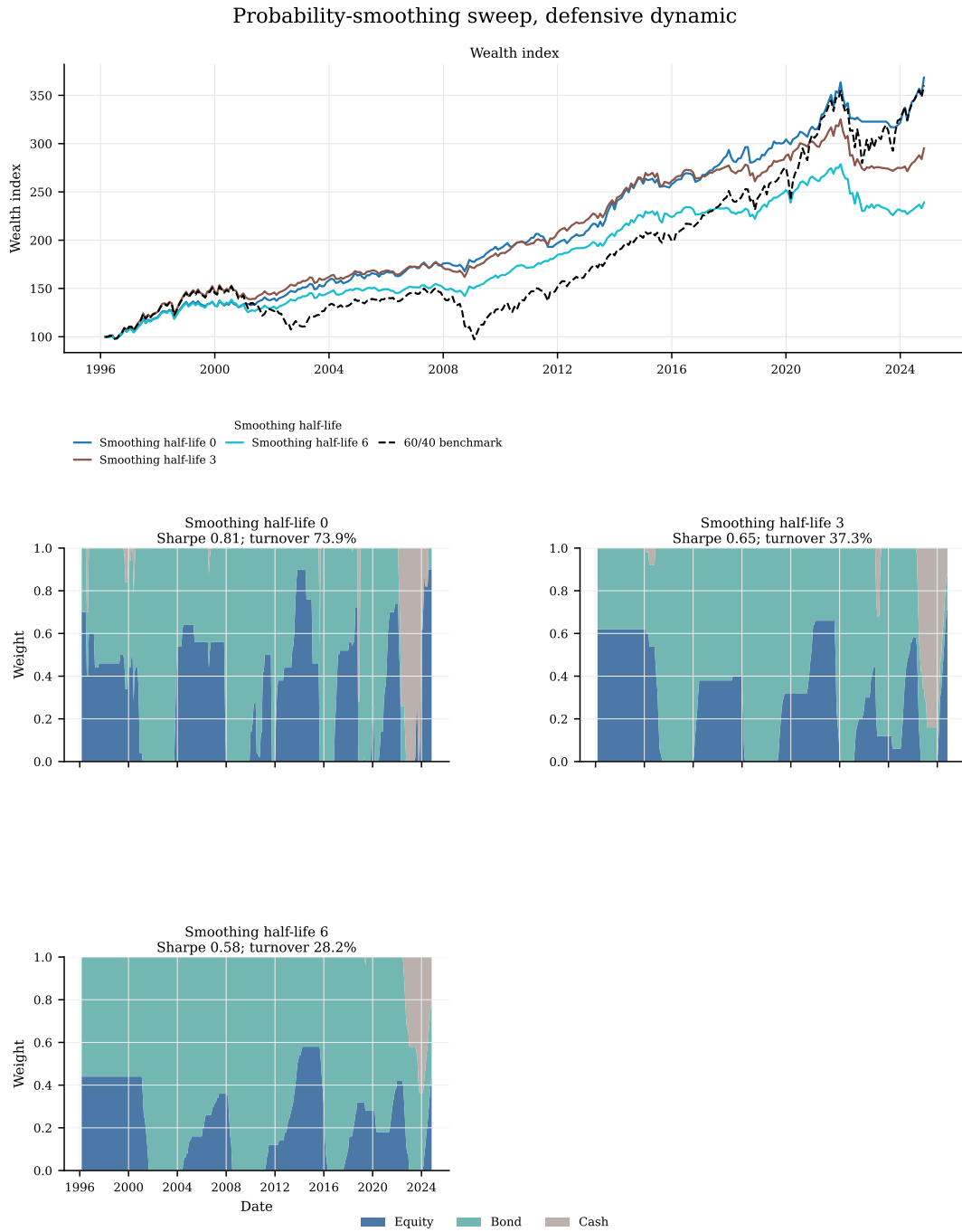


Figure 8: Probability-smoothing sweep, defensive dynamic

Note. Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table 2.

Table 7: Probability-smoothing sweep, defensive dynamic

Smoothing half-life	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	4.7%	5.8%	0.81	-12.8%	-5.2%	368.53	73.9%
3	4.0%	6.1%	0.65	-16.6%	-6.0%	295.10	37.3%
6	3.2%	5.5%	0.58	-19.0%	-5.3%	239.07	28.2%



Figure 9: Probability-smoothing sweep, low-turnover dynamic

Note. Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table 2.

Table 8: Probability-smoothing sweep, low-turnover dynamic

Smoothing half-life	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	5.9%	8.0%	0.74	-11.9%	-7.7%	495.82	196.9%
3	5.3%	7.2%	0.74	-12.2%	-7.2%	426.74	57.3%
6	4.3%	6.9%	0.62	-14.8%	-6.7%	322.55	38.9%

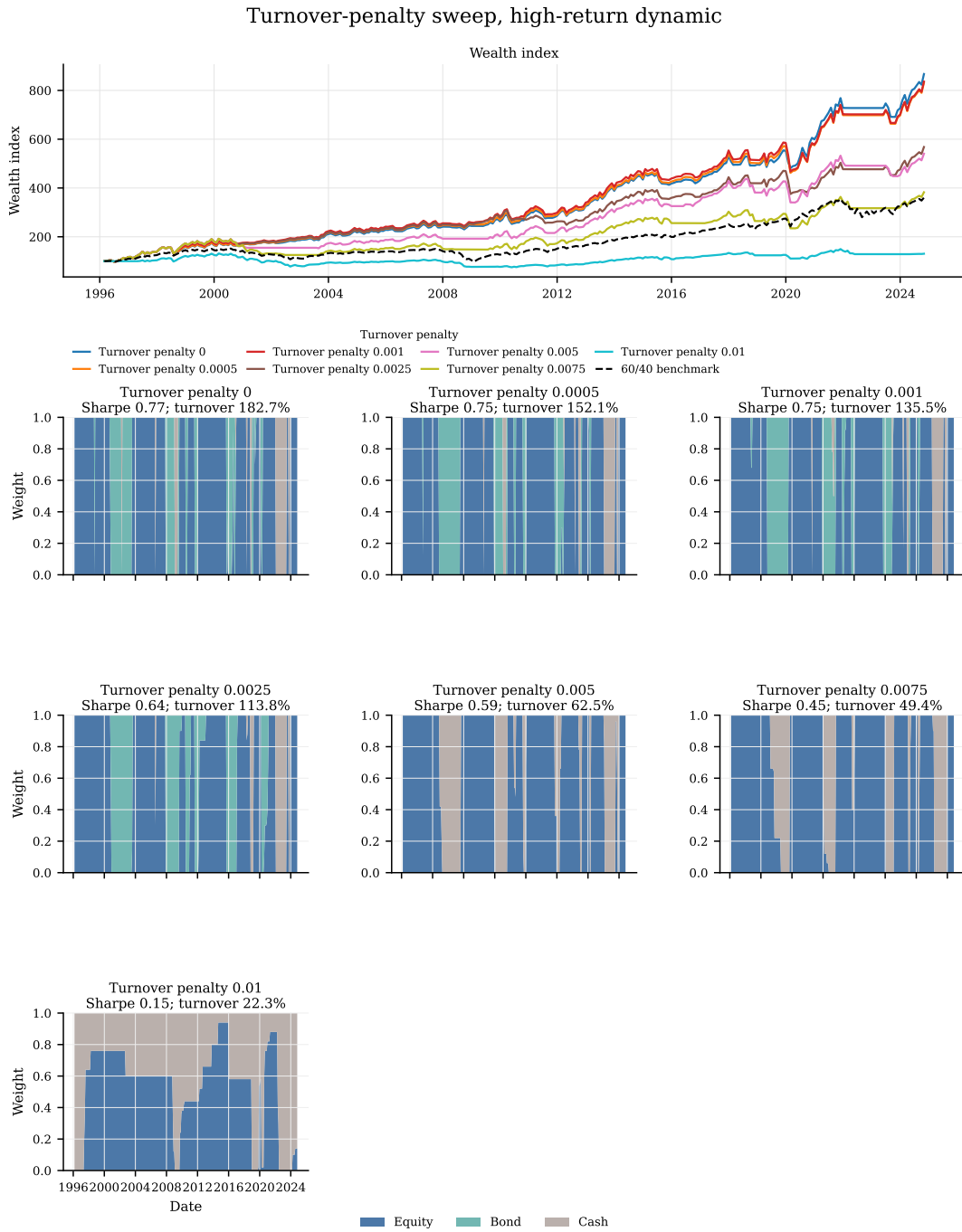


Figure 10: Turnover-penalty sweep, high-return dynamic

Note. Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table 2.

Table 9: Turnover-penalty sweep, high-return dynamic

Turnover penalty	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	8.1%	10.4%	0.77	-16.3%	-9.7%	867.20	182.7%
0.0005	8.0%	10.7%	0.75	-19.2%	-10.8%	831.93	152.1%
0.001	8.0%	10.7%	0.75	-20.3%	-11.1%	836.31	135.5%
0.0025	6.6%	10.3%	0.64	-20.3%	-11.1%	568.07	113.8%
0.005	6.5%	11.1%	0.59	-22.1%	-11.2%	540.65	62.5%
0.0075	5.4%	11.9%	0.45	-34.7%	-11.7%	382.60	49.4%
0.01	1.3%	9.0%	0.15	-43.1%	-9.5%	130.50	22.3%

Note. Higher turnover penalties reduce trading and increase cash weights.



Figure 11: Turnover-penalty sweep, defensive dynamic

Note. Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table 2.

Table 10: Turnover-penalty sweep, defensive dynamic

Turnover penalty	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	4.9%	6.5%	0.75	-11.2%	-6.4%	380.55	144.3%
0.0005	4.8%	6.1%	0.79	-11.8%	-5.9%	376.11	98.4%
0.001	4.7%	5.8%	0.81	-12.8%	-5.2%	368.53	73.9%
0.0025	3.7%	6.0%	0.63	-17.8%	-6.2%	277.40	39.0%
0.005	2.2%	5.7%	0.38	-16.9%	-6.3%	178.39	23.3%
0.0075	0.5%	3.2%	0.17	-19.1%	-3.8%	114.78	9.3%
0.01	0.1%	0.9%	0.15	-6.7%	-1.1%	104.02	0.4%

Note. Higher turnover penalties reduce trading and increase cash weights.



Figure 12: Turnover-penalty sweep, low-turnover dynamic

Note. Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table 2.

Table 11: Turnover-penalty sweep, low-turnover dynamic

Turnover penalty	Ann. return	Ann. vol.	Sharpe	Max. drawdown	Monthly CVaR 1%	Final wealth	Ann. turnover
0	5.2%	7.3%	0.71	-12.0%	-6.8%	416.60	93.6%
0.0005	5.2%	7.4%	0.71	-12.7%	-7.1%	412.82	74.8%
0.001	5.3%	7.2%	0.74	-12.2%	-7.2%	426.74	57.3%
0.0025	4.3%	6.9%	0.62	-15.5%	-6.8%	322.13	34.9%
0.005	2.9%	6.6%	0.43	-23.0%	-6.8%	214.41	18.9%
0.0075	1.6%	4.7%	0.34	-22.8%	-4.9%	152.19	8.3%
0.01	0.3%	1.5%	0.23	-7.7%	-1.5%	110.14	1.1%

Note. Higher turnover penalties reduce trading and increase cash weights.

Table 12: Transaction-cost and turnover-penalty sensitivity

Strategy	Turnover penalty	5 bps	10 bps	25 bps
High-return dynamic	0	0.77	0.74	0.67
	0.0005	0.75	0.74	0.62
	0.001	0.75	0.72	0.53
	0.0025	0.64	0.58	0.44
	0.005	0.59	0.56	0.47
	0.01	0.37	0.17	
Defensive dynamic	0	0.81	0.78	0.69
	0.0005	0.79	0.81	0.61
	0.001	0.81	0.77	0.57
	0.0025	0.63	0.58	0.43
	0.005	0.38	0.35	0.20
	0.01	0.15	0.35	
Low-turnover dynamic	0	0.74	0.70	0.63
	0.0005	0.71	0.73	0.61
	0.001	0.74	0.69	0.59
	0.0025	0.62	0.60	0.50
	0.005	0.43	0.43	0.38
	0.01	0.23	0.33	

Note. Entries are Sharpe ratios. For each dynamic strategy, the model, predictor set, risk aversion, and smoothing half-life are fixed at the strategy-specific baseline values in Table 2.